

EXAMPLE 3**Republic of India Semiannual Coupon Bond**

Consider the 20-year Republic of India government bond from which the discount bond in Example 1 was separated (or stripped). The bond was issued at an annualized coupon rate of 6.70 percent and a YTM of 6.70 percent, and the coupon payments are semiannual.

1. Solve for the price of the bond at issuance.

Solution:

INR100

As coupon periods are semiannual, for a principal of INR100, $PMT = \text{INR}3.35 (=6.70/2)$ and the periodic discount rate is 3.35 percent $(=6.70 \text{ percent}/2)$, as follows:

$$PV = 3.35/(1.0335) + 3.35/(1.0335)^2 + \dots + 3.35/(1.0335)^{39} + 103.35/(1.0335)^{40}.$$

As shown in Example 2, because *the coupon rate is equal to the YTM*, we expect this bond to have a PV of INR100 at issuance.

2. What is the bond's price if the YTM immediately rises to 7.70 percent?

Solution:

INR89.88

We can solve for the PV as INR89.88 using Equation 6 with $r = 3.85\%$ $(=7.70/2)$ or using the Microsoft Excel or Google Sheets PV function ($PV(0.0385, 40, 3.35, 100, 0)$). The first and final three cash flows are shown below:

		Years	0	0.5	1	1.5	19	19.5	20
r		Periods	0	1	2	3	38	39	40
	FV			3.35	3.35	3.35	3.35	3.35	103.35
6.70%	PV		100.00	3.24	3.14	3.03	0.96	0.93	27.66
7.70%	PV		89.88	3.23	3.11	2.99	0.80	0.77	22.81

3. Recalculate the discount bond price for the final principal payment in 20 years from Example 1 using a 6.70 percent semiannual discount rate.

Solution:

INR26.77

Note that the PV calculation using the same annual discount rate for 40 semiannual periods will differ slightly using Equation 5, as follows:

$$PV = \text{INR}27.66 = (PMT_{40} + FV_{40})/(1+r/2)^{40},$$

$$PV(PMT_{40}) = \text{INR}0.90 = 3.35 / (1.0335)^{40},$$

$$PV(FV_{40}) = \text{INR}26.77 = 100 / (1.0335)^{40}.$$

Compounding on a semiannual basis for 40 periods, $PV(FV_{40})$ of 26.77 is less than the original PV of 27.33 using 20 annual periods from Example 1 (since $1/(1+r)^t > 1/(1+r/2)^{2t}$ when $r \geq 0$).